Exercises 6

Exercise 1 (Hadamard transform)**.** *Recall that $s \cdot x = \sum_{i=1}^{n} s_i x_i$ denotes the inner product of bit strings $s, x \in \{0,1\}^n$ modulo 2.*

1. *Verify that, for any $s \in \{0,1\}^n$,*

$$\frac{1}{\sqrt{2}}\Big(|0\rangle + (-1)^{s_1}|1\rangle\Big) \otimes \cdots \otimes \frac{1}{\sqrt{2}}\Big(|0\rangle + (-1)^{s_n}|1\rangle\Big) = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x}|x\rangle.$$

2. *Let $s \in \{0,1\}^n$. Verify that*

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} = \delta_{s,0^n} = \begin{cases} 1 & \text{if } s = 0 \ldots 0, \\ 0 & \text{otherwise.} \end{cases}$$

3. *Recall that the entry of $H^{\otimes n}$ at row $x \in \{0,1\}^n$ and column $y \in \{0,1\}^n$ is*

$$\frac{1}{\sqrt{2^n}}(-1)^{x \cdot y}.$$

Using this, verify explicitly that $H^{\otimes n}$ is a unitary matrix, i.e., that $H^{\otimes n}(H^{\otimes n})^\dagger = I$.

Exercise 2 (Deterministic algorithm for Bernstein–Vazirani)**.** *Find a deterministic classical algorithm that solves the Bernstein–Vazirani problem with certainty by evaluating the function exactly n times. In other words, find a collection of n strings $x \in \{0,1\}^n$ such that evaluating the function $f(x) = s \cdot x$ on them uniquely determines s.*

Exercise 3 (Bernstein–Vazirani for an affine function)**.** *Consider a slight extension of the Bernstein–Vazirani problem where the function f within the oracle is affine rather than linear, i.e., $f(x) = s \cdot x + b$, for some unknown $s \in \{0,1\}^n$ and $b \in \{0,1\}$. Would the final measurement still return $s \in \{0,1\}^n$ correctly?*

Exercise 4 (All ones in Deutsch–Jozsa)**.** *When deriving the Deutsch–Jozsa algorithm, we looked at the probability of getting all zeroes in the final measurement. Let's say we instead looked at the probability of getting all ones. Characterize those Boolean functions $f : \{0,1\}^n \to \{0,1\}$ for which the final measurement produces the outcome $z = 1^n = 11 \ldots 1$ with probability 1.*

Exercise 5 (`Cirq`: Bernstein–Vazirani)**.**

Exercise 6 (`Cirq`: Deutsch–Jozsa)**.**